

Statistical Quality Control

Pareto Diagram

A Pareto diagram, developed by Italian economist **Vilfredo Pareto**, is a problem-solving tool that helps prioritize efforts to meet customer needs. It combines a vertical **bar chart** and a **line graph**, with bars arranged left to right in descending order of frequency. The x-axis shows the independent variable, while bar length represents frequency.

The principle behind Pareto analysis is that roughly **80% of problems arise from 20% of causes**. By focusing on these major causes, most issues can be resolved. The chart highlights the most significant data points, enabling targeted improvements.

Pareto charts are widely used in quality control to:

1. Arrange data in descending order.
2. Identify priority areas for improvement.

Utilization of Pareto Diagram

1. Identification of the most significant problem.
2. Identification of the reason for the problem.
3. To prioritize actions.
4. To review the corrective measure that has been taken.
5. Allows explaining of important tasks.

Advantages of Pareto Analysis

1. It is an easy and efficient tool of quality control.
2. The Pareto chart provide the basis for the segregation of a problem and its root cause.
3. It helps to identify the minimum number of problems that creates the maximum number of problems.

Limitations of Pareto Analysis

1. The Pareto principle is not a universal rule; it cannot be applicable in all cases.
2. It is unable to provide root cause of problem; it only shows the major problems.
3. For large volumes of data, it is sometime difficult to draw the Pareto chart effectively.
4. A Pareto chart is only focused on past value or data. It is not useful for future or present/current scenario

Organize Data

- i. Collect data
- ii. Create a data table
- iii. Arrange the data in descending order
- iv. Sum the totals
- v. Calculate individual percentages
- vi. Calculate cumulative percentages
- vii. *Plot the Diagram***
 - a) Step 1: Select the items for which the Pareto analysis is required.
 - b) Step 2: Draw the horizontal and vertical lines showing the x axis and the y axis.
 - c) Step 3: Divide the x axis and the y axis on a suitable scale as per the requirement.
 - d) Step 4: Draw the bars from the extreme left side of highest frequency and move to the right side with decreasing frequency.
 - e) Step 5: Labels the bars. On the horizontal axis below each bar, label the bars so as to know which cause each bar represents, as shown in the graph below.
 - f) Step 6: Continue in this way to make all the bars.

- g) Step 7: Make the dot for each and every item, corresponding to the frequency for each item. For every item, plot a dot on the graph. For this dot, start from the left, point on the bar or above it, corresponding to the value of frequencies for each of them.
- h) Step8: Join all the dots by using a scale.
- i) Step 9: Mention the title of the chart.

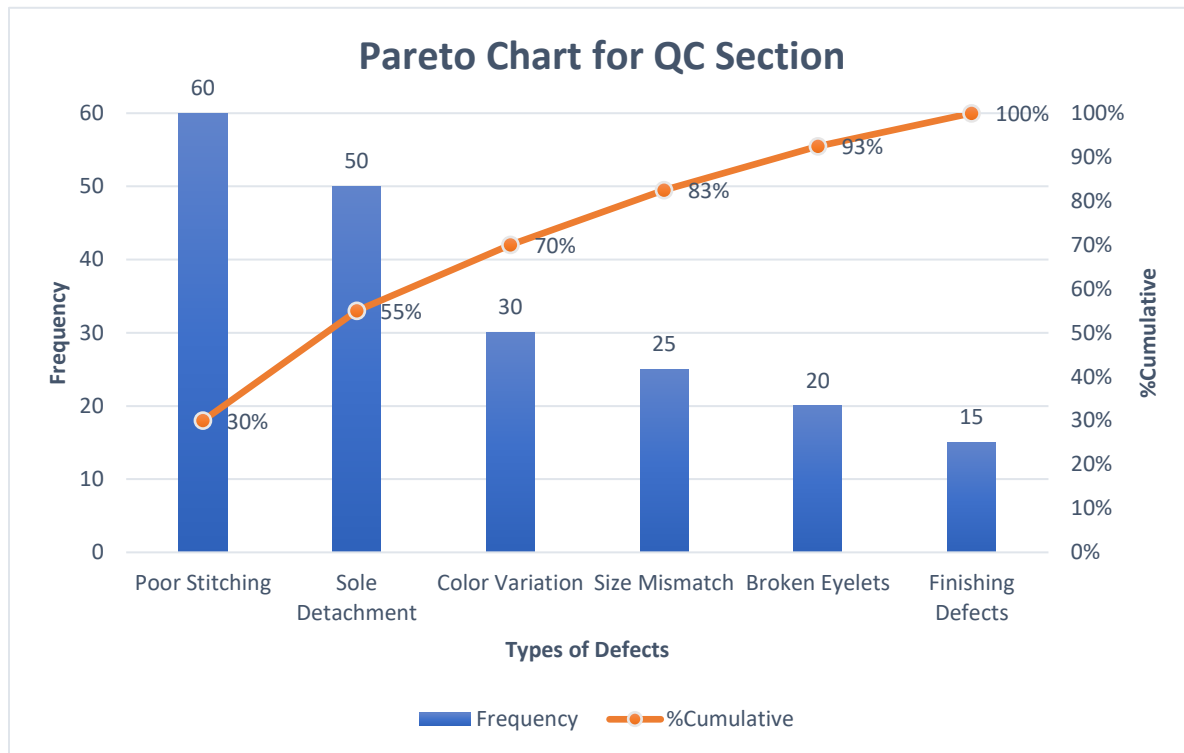
Case Study: Reducing Defects in Footwear Manufacturing Using Pareto Analysis

Background

A footwear manufacturing company noticed a high rate of customer complaints about defective shoes. Management wanted to identify the most common types of defects and prioritize improvements using Pareto analysis.

Problem Statement: The company received 200 defect reports over one month. The quality control team classified the defects into six categories:

Defect Type	Frequency
Poor Stitching	60
Sole Detachment	50
Color Variation	30
Size Mismatch	25
Broken Eyelets	20
Finishing Defects	15



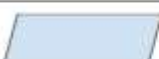
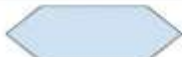
Process Flow Diagram

A TQM (Total Quality Management) process flow diagram is a visual representation of business and industrial processes, often using a flowchart format, to understand, analyze, and improve quality-related operations to ensure customer satisfaction. It maps out key steps and identifies areas for quality improvement by highlighting the flow of activities, decision points, and potential issues like delays or errors. These diagrams are created with the help of specialized software or templates and serve as a communication tool to facilitate understanding and collaboration among team members.

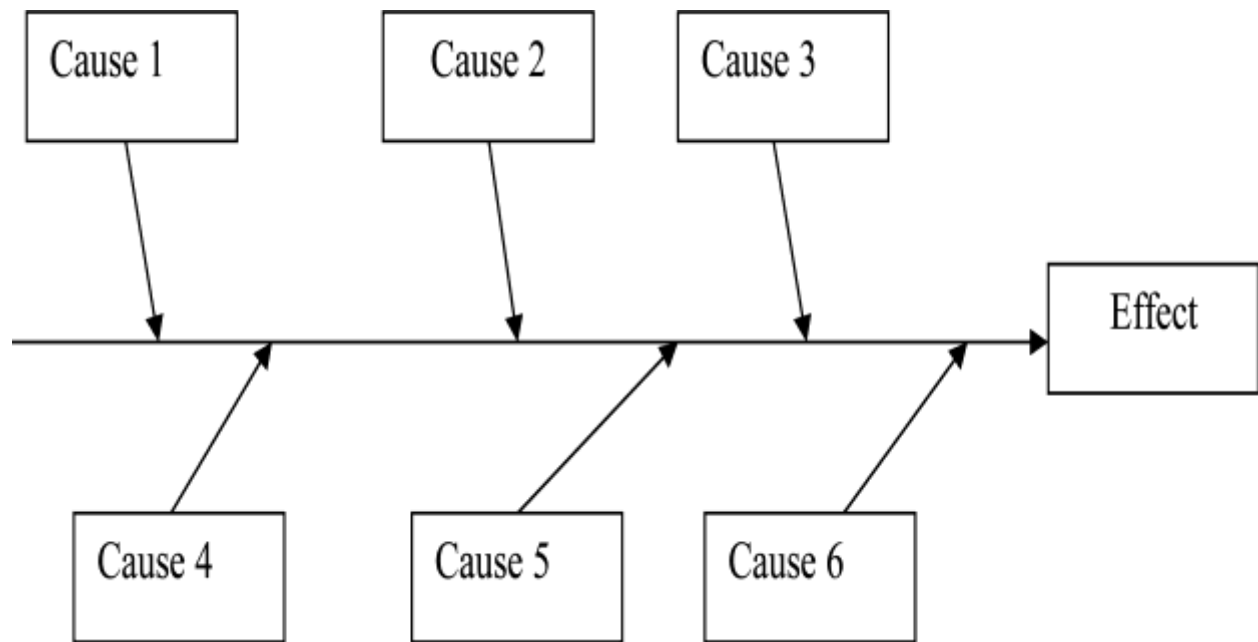
Purpose of a TQM Process Flow Diagram

- ☐ Provides a clear, step-by-step understanding of a process, making it easier for everyone involved to grasp.
- ☐ Helps in identifying areas for improvement, such as bottlenecks, sources of error, duplication, or unnecessary steps.

- ❑ Facilitates better communication and teamwork by providing a common visual reference point for process analysis.
- ❑ Focuses on improving processes to enhance customer satisfaction by ensuring quality standards are met.
- ❑ By streamlining processes and reducing errors, it can contribute to more cost-efficient operations.

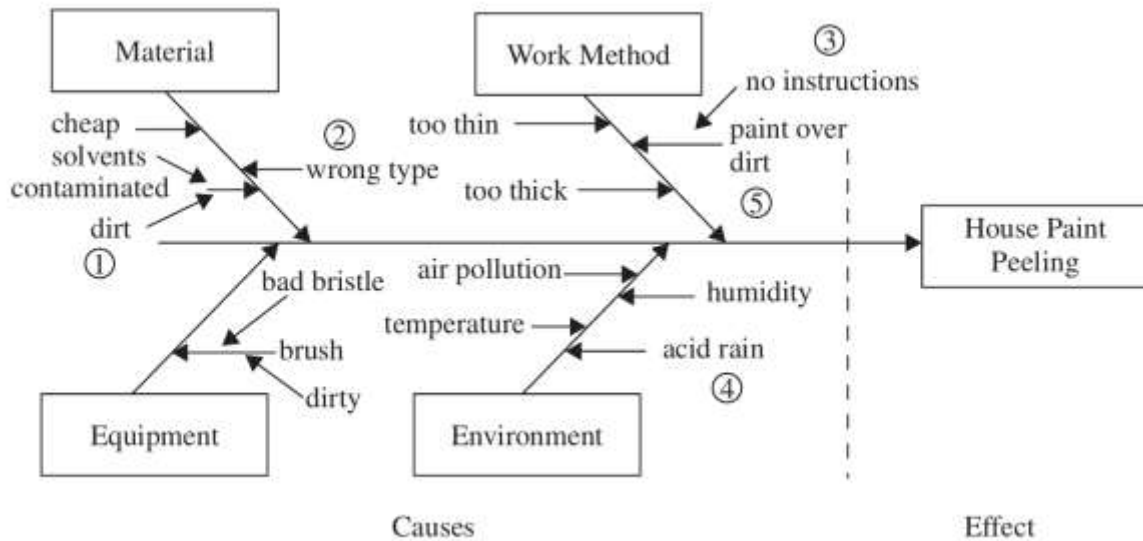
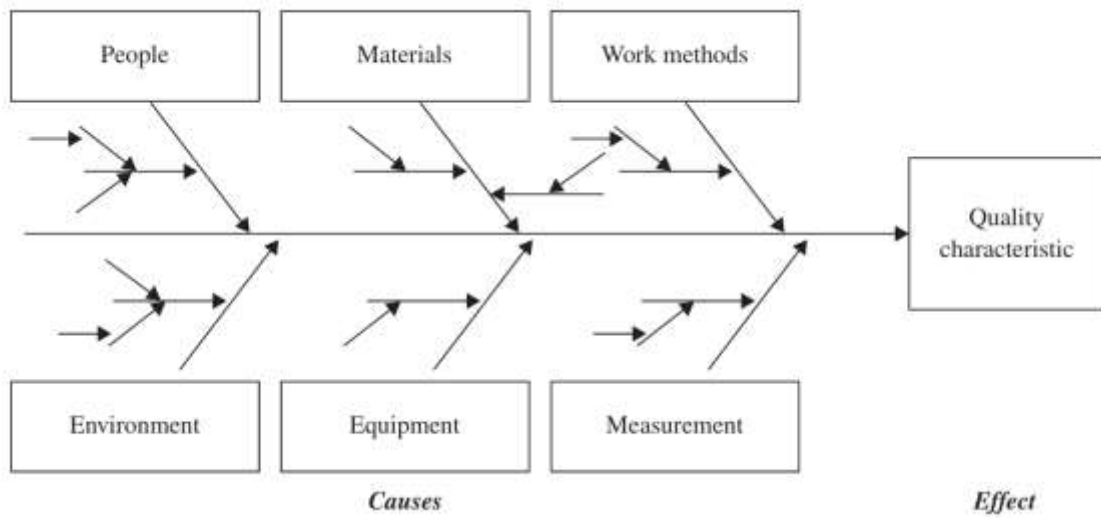
Symbol	Name	Function
	Start/End	Indicates the start or end point of the process.
	Process	Describes a specific action or task to be performed within the process.
	Decision	Denotes a question or condition to be evaluated. Also, the different output options based on the answer.
	Connector	Used to connect different parts of the diagram that are in separate places.
	Flow arrow	It represents the direction of the process flow, indicating the order in which actions are carried out.
	Input/Output	It is the material or data entering or leaving the system.
	Document	Refers to an external document or file
	Multi-document	Indicates several printed documents or reports
	Preparation	Marks an adjustment to another step in the process.

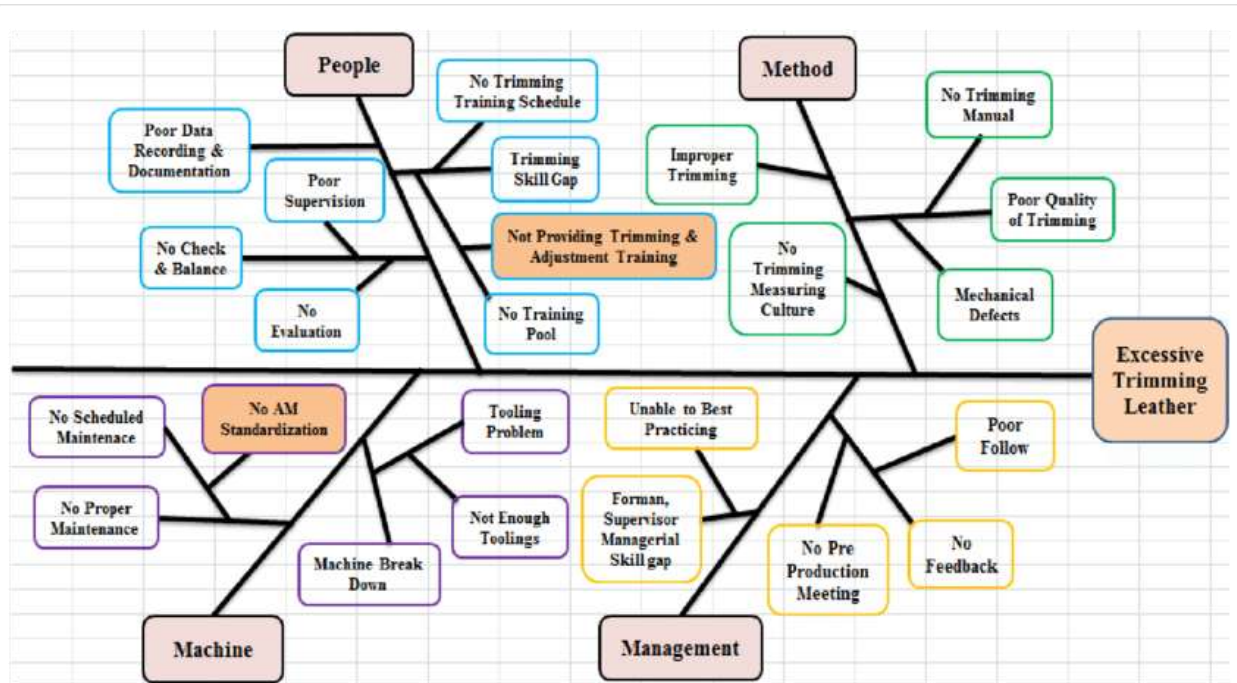
Cause-and-Effect Diagram/ Fishbone Diagram



The diagrams are useful to

1. Analyze actual conditions for the purpose of product or service quality improvement, more efficient use of resources, and reduced costs.
2. Eliminate conditions causing nonconformities and customer complaints.
3. Standardize existing and proposed operations.
4. Educate and train personnel in decision-making and corrective-action activities



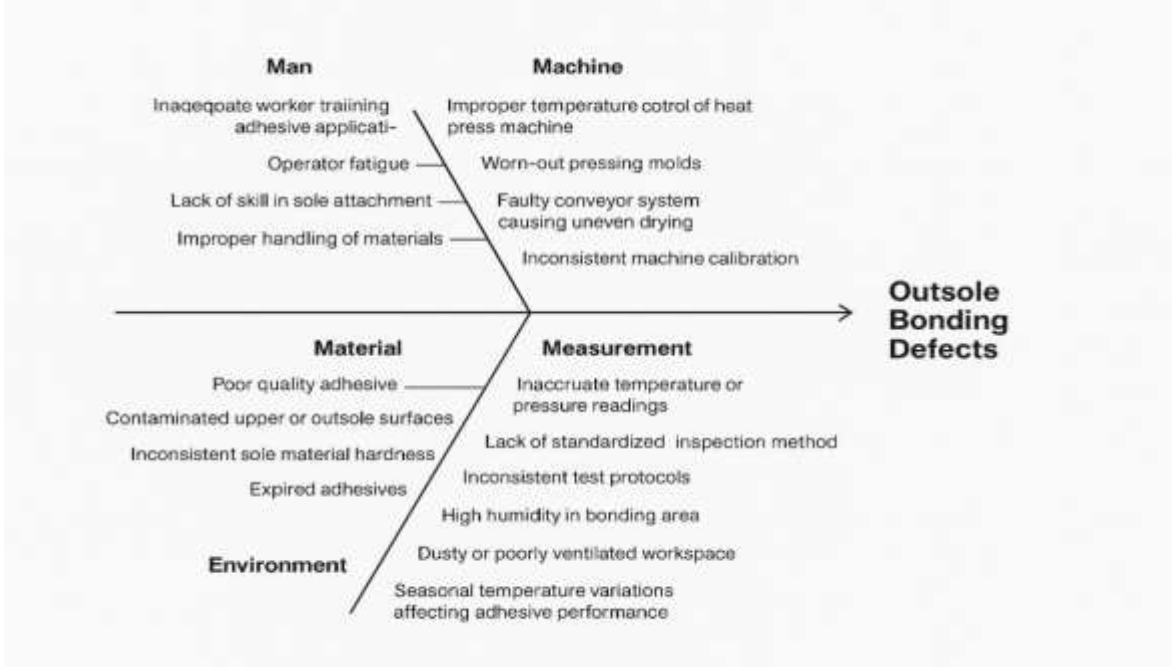


Case Study: Cause-and-Effect Diagram

A mid-sized footwear manufacturing company, XYZ Footwear Ltd., specializes in producing sports and casual shoes for export. Recently, the quality control (QC) team observed a rise in outsole bonding failures during final inspections. Shoes from multiple production lines showed detachment of soles after flexing and wear tests, leading to increased rejection rates (8% compared to the usual 3%). The company decided to use a Cause and Effect (Fishbone) Diagram to systematically identify possible causes and take corrective actions

Man (People)	- Inadequate worker training on adhesive application - Operator fatigue - Lack of skill in sole attachment - Improper handling of materials
Machine	- Improper temperature control of heat press machine - Worn-out pressing molds - Faulty conveyor system causing uneven drying - Inconsistent machine calibration

Material	- Poor quality adhesive - Contaminated upper or outsole surfaces - Inconsistent sole material hardness - Expired adhesives
Method (Process)	- Incorrect adhesive mixing ratio - Inadequate drying time before pressing - Poor surface preparation (dust, oil, moisture) - Skipping quality checks
Measurement	- Inaccurate temperature or pressure readings - Lack of standardized inspection method - Inconsistent test protocols
Environment	- High humidity in bonding area - Dusty or poorly ventilated workspace - Seasonal temperature variations affecting adhesive performance



Control Chart

A control chart is a statistical process control tool used to monitor process stability and determine whether a process is under control. It consists of three key lines: a center line (CL), an upper control limit (UCL), and a lower control limit (LCL), plotted over time to analyze process behavior.

Control limits are based on historical data showing the process's central tendency and variation. If measurements fall within these limits, the process is considered in control; if not, corrective actions are needed.

Variations in a process are due to:

- **Chance (random) causes:** Minor, unavoidable factors (e.g., air friction, temperature changes).
- **Assignable causes:** Major, controllable factors (e.g., raw material quality, worker skills, tools).

Purpose of control charts:

- Detect when a process is out of control.
- Identify and correct assignable causes.
- Maintain consistent product quality and meet customer requirements.

Control Charts



Three Sigma Capability

- Mean output ± 3 standard deviations falls within the design specification
- It means that 0.26% of output falls outside the design specification and is unacceptable
- The result: a 3-sigma capable process produces 2600 defects for every million units produced

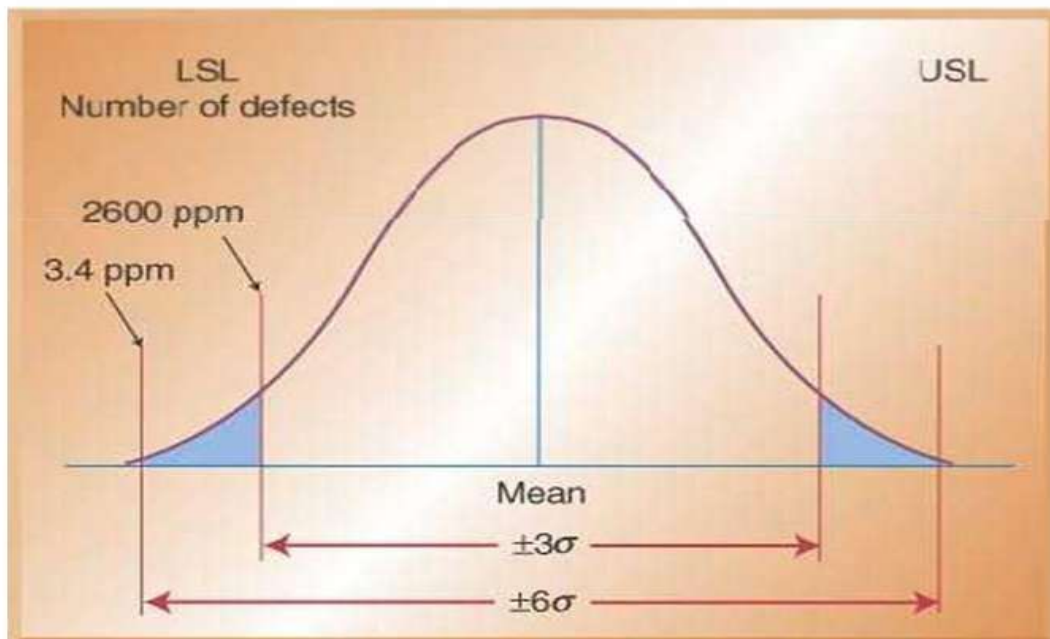
Control Charts

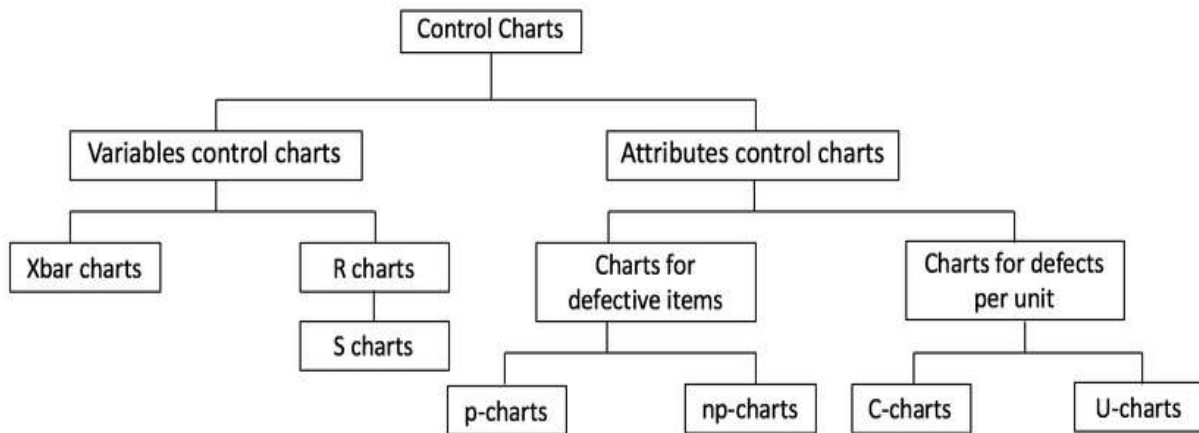


Six Sigma Capability

- Six sigma capability assumes the process is capable of producing output where the mean ± 6 standard deviations fall within the design specifications
- The result: only 3.4 defects for every million produced
- Six sigma capability means smaller variation and therefore higher quality

Three/Six Sigma Capability





Steps to Plot X and R Chart

Step 1 – Find the mean value ($\bar{X}$). Use the following formula for each subgroup

Step 2 – Find the range, $R = X(\text{maximum value}) - X(\text{minimum value})$

Step 3 – Find the overall mean, or $\bar{\bar{X}}$.

Step 4 – Compute the average value of the range ($\bar{R}$).

Step 5 – Compute the Control Limit Lines.

X Control Chart

$$\text{Central Limit (CL)} = \bar{\bar{X}}$$

$$\text{Upper Control Limit (UCL)} = \bar{\bar{X}} + A_2 * \bar{R}$$

$$\text{Lower Control Limit (LCL)} = \bar{\bar{X}} - A_2 * \bar{R}$$

R Control Chart

$$\text{Central Line (CL)} = \bar{R}$$

Factors for Computing Central Lines and 3 σ Control Limits for Variables Charts

Sample Size	CHART FOR AVERAGES		CHART FOR STANDARD DEVIATIONS		
	Factors for Control Limits		Factor for Central Line	Factors for Control Limits	
	A_2	A_3	C_4	B_3	B_4
2	1.880	2.659	0.7979	0	3.267
3	1.023	1.954	0.8862	0	2.568
4	0.729	1.628	0.9213	0	2.266
5	0.577	1.427	0.9400	0	2.089
6	0.483	1.287	0.9515	0.030	1.970
7	0.419	1.182	0.9594	0.118	1.882
8	0.373	1.099	0.9650	0.185	1.815

Sample Size	CHART FOR RANGES				
	Factor for Central Line	Factors for Control Limits			
	d_2	D_3	D_4	D_5	D_6
2	1.128	0	3.267	0	3.865
3	1.693	0	2.574	0	2.745
4	2.059	0	2.282	0	2.375
5	2.326	0	2.114	0	2.179
6	2.534	0	2.004	0	2.055
7	2.704	0.076	1.924	0.078	1.967
8	2.847	0.136	1.864	0.139	1.901

Problem:

Data on the Depth of the Shoe Sole (millimeters)

Plot the control limits for $\bar{X}$ bar and R charts for the given data

<i>Subgroup Number</i>	<i>Date</i>	<i>Time</i>	<i>X₁</i>	<i>X₂</i>	<i>X₃</i>	<i>X₄</i>
<i>1</i>	<i>23-Jul</i>	<i>8:50</i>	<i>35</i>	<i>40</i>	<i>32</i>	<i>37</i>
<i>2</i>		<i>11:30</i>	<i>46</i>	<i>37</i>	<i>36</i>	<i>41</i>
<i>3</i>		<i>1:45</i>	<i>34</i>	<i>40</i>	<i>34</i>	<i>36</i>
<i>4</i>		<i>3:45</i>	<i>69</i>	<i>64</i>	<i>68</i>	<i>59</i>
<i>5</i>		<i>4:20</i>	<i>38</i>	<i>34</i>	<i>44</i>	<i>40</i>
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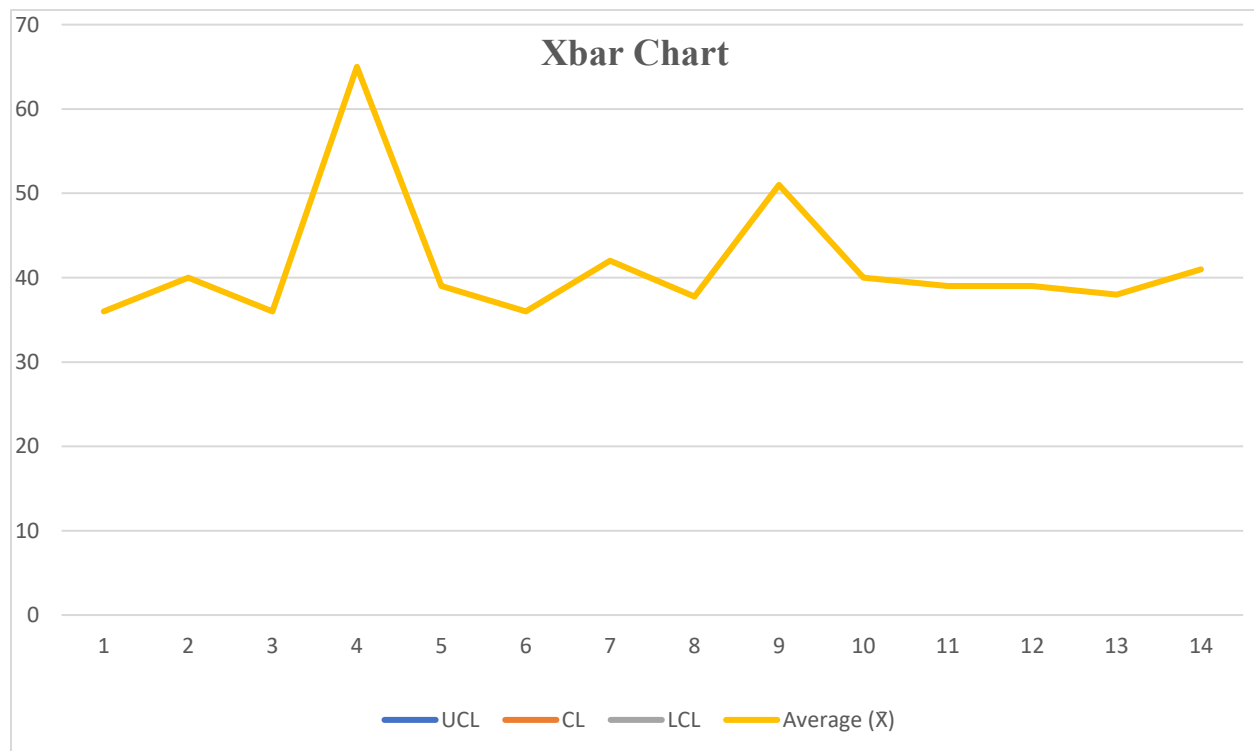
$$\text{Upper Control Limit (UCL)} = D4 * \bar{R}$$

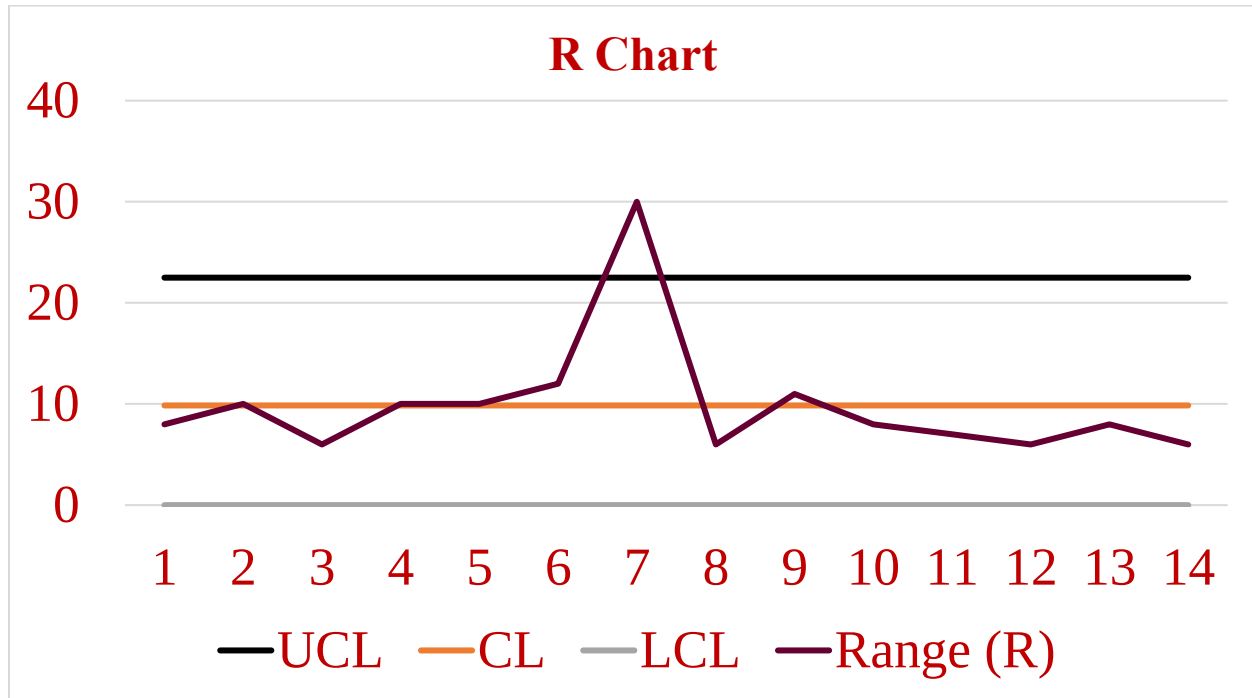
$$\text{Lower Control Limit (LCL)} = D3 * \bar{R}$$

Solution:

Subgroup Number	X1	X2	X3	X4	Average ($\bar{X}$)	Range (R)
1	35	40	32	37	36	8
2	46	37	36	41	40	10
3	34	40	34	36	36	6
4	69	64	68	59	65	10
5	38	34	44	40	39	10
6	41	40	29	34	36	12
7	38	44	28	58	42	30
8	35	41	37	38	37.75	6
9	56	55	45	48	51	11
10	38	40	45	37	40	8
11	39	42	35	40	39	7
12	42	39	39	36	39	6
13	43	36	35	38	38	8
14	39	38	43	44	41	6
Xbar					41.41071429	
R bar					9.857142857	

Xbar			R bar		
UCL	CL	LCL	UCL	CL	LCL
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0





Process Capability

Process capability measures how well an in-control process can consistently meet customer-specified tolerance limits. This analysis helps identify process variability, predict process sufficiency, and ensure consistency. By understanding process capability, organizations can set realistic specifications, plan processes effectively, choose reliable suppliers, and reduce variability, leading to fewer defects and better-quality outcomes.

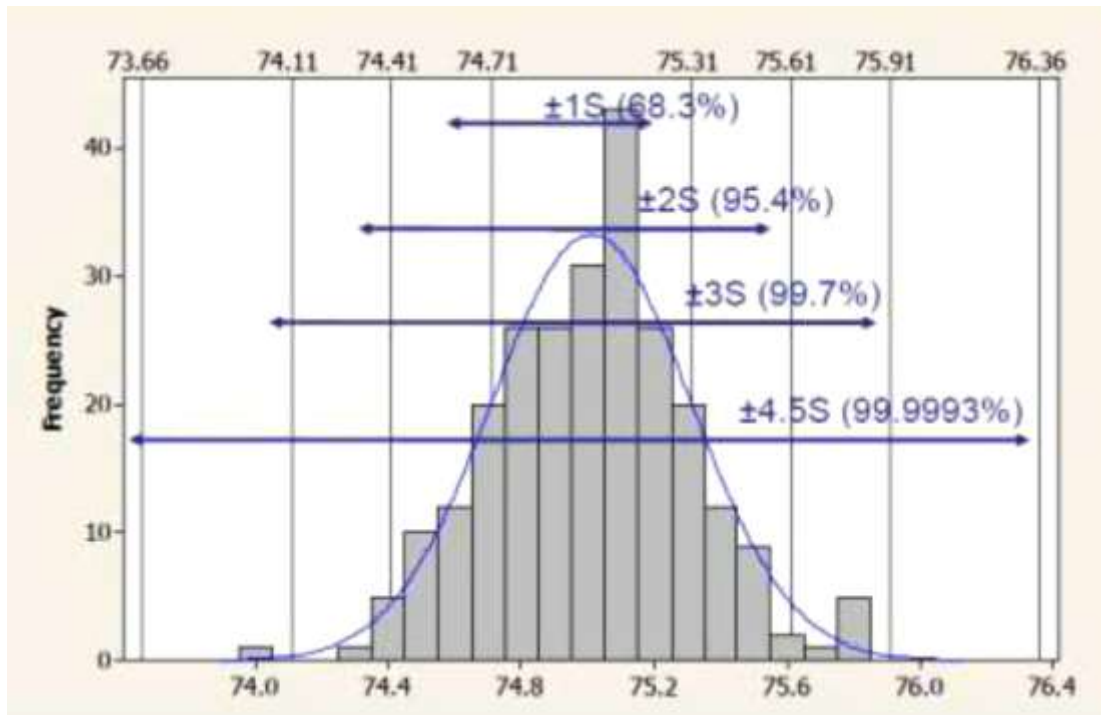
$$\text{Process Capability Ratio (PCR)} = Cp = (USL - LSL) / 6\sigma$$

Where,

USL = upper specification limit

LSL = lower specification limit

σ = process standard deviation



Capability Index (Cp)

- $C_p > 1.00 \rightarrow$ Process is capable of meeting specifications.
- $C_p < 1.00 \rightarrow$ Process is not capable of meeting specifications.

Industry Standards

- $C_p = 1.33 \rightarrow$ Commonly accepted standard.
- $C_p = 2.00 \rightarrow$ Higher benchmark used by some organizations.
- Achieving a higher C_p requires realistic specification limits and continuous improvement of process capability. However, C_p does not reflect how well the process is centered relative to the target value; this is measured by **Cpk**.
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$$C_{pk} = \frac{\text{Min} \{ (\text{USL} - \bar{X}) \text{ or } (\bar{X} - \text{LSL}) \}}{3\sigma}$$

- $C_{pk} = 1.00 \rightarrow$ De facto standard.
- $C_{pk} = 1.33 \rightarrow$ Used by some organizations.

$$\sigma = \frac{r}{d_2}$$

- $C_{pk} = 1.00 \rightarrow$ De facto standard.
- $C_{pk} = 1.33 \rightarrow$ Used by some organizations

Some points to be noted,

1. The C_p value does not change as the process center changes.
2. $C_p = C_{pk}$ when the process is centered.
3. C_{pk} is always equal to or less than C_p .
4. A C_{pk} value greater than 1.00 indicates the process conforms to specifications.
5. A C_{pk} value less than 1.00 indicates that the process does not conform to specifications.
6. A C_p value less than 1.00 indicates that the process is not capable.
7. A C_{pk} value of zero indicates the average is equal to one of the specification limits.

8. A negative C_{pk} value indicates that the average is outside the specifications.

Sample Size	CHART FOR RANGES					Chart for Medians
	Factor for Central Line	Factors for Control Limits				
		D_3	D_4	D_5	D_6	
n	d_2					A_5
2	1.128	0	3.267	0	3.865	2.224
3	1.693	0	2.574	0	2.745	1.265
4	2.059	0	2.282	0	2.375	0.829
5	2.326	0	2.114	0	2.179	0.712
6	2.534	0	2.004	0	2.055	0.562
7	2.704	0.076	1.924	0.078	1.967	0.520
8	2.847	0.136	1.864	0.139	1.901	0.441

Problem

The specifications for one characteristic of a part call for its width to be 3.000 ± 0.008 cm. The process has been run under controlled conditions so that no assignable causes of variation have been introduced, and samples have been taken. The standard deviation of the process was estimated to be 0.003 cm.

1. Cp What is Cp for this process? What does this value say about the process capability?
2. What percentage of the units would be outside the specifications for the width of the part if the process were operated and centered at 3.000 centimeters? Assume a normal distribution of measurements.

Problem

<i>Subgroup Number</i>	<i>Date</i>	<i>Time</i>	<i>X₁</i>	<i>X₂</i>	<i>X₃</i>	<i>X₄</i>
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Compute the C_p and C_{pk} from the following data table

Sampling Plans

Three often-used attribute sampling plans are

1. Single-sampling plan
2. Double-sampling plan
3. Sequential-sampling plan

Single-sampling plan

The single-sampling plan is a decision rule to accept or reject a lot based on the results of one random sample from the lot. The procedure is to take a random sample of size (n) and inspect each item. If the number of defects does not exceed a specified acceptance number (c), the consumer accepts the entire lot. Any defects found in the sample are either repaired or returned to the producer. If the number of defects in the sample is greater than c , the consumer subjects the entire lot to 100 percent inspection or rejects the entire lot and returns it to the producer.

Double-Sampling Plan

In a double-sampling plan, management specifies two sample sizes (n_1 and n_2) and two acceptance numbers (c_1 and c_2).

Procedure:

Take first sample of size n_1 .

- If defectives $\leq c_1 \rightarrow$ **Accept lot.**
- If defectives $> c_2 \rightarrow$ **Reject lot.**
- If $c_1 < \text{defectives} \leq c_2 \rightarrow$ Take second sample of size n_2 .
- If total defectives ($n_1 + n_2$) $\leq c_2 \rightarrow$ **Accept lot.**
- Otherwise $\rightarrow$ **Reject lot.**

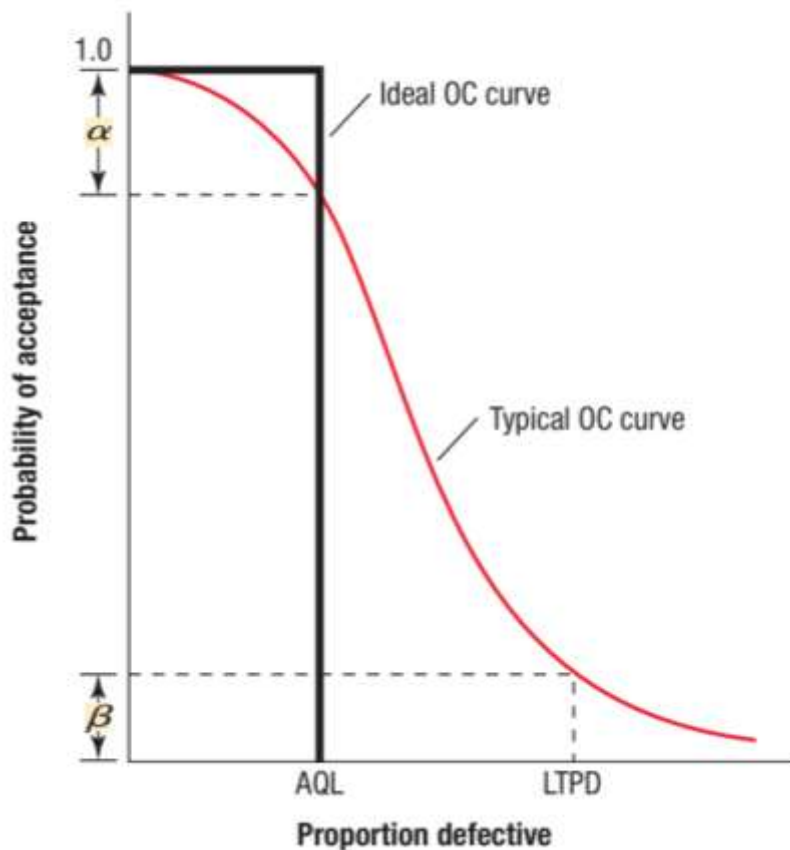
Advantages

- Saves inspection cost when lots are very good or very bad (decision made after first sample).
- Provides flexibility compared to single-sampling.

Disadvantage: If a second sample is needed, inspection cost may exceed that of a single-sampling plan.

Operating Characteristic (OC) Curves

Analysts create a graphic display of the performance of a sampling plan by plotting the probability of accepting the lot for a range of proportions of defective units. This graph, called an operating characteristic (OC) curve, describes how well a sampling plan discriminates between good and bad lots.



1. It is a graph of the % defective (p) in a lot or batch vs. the probability that the sampling plan will accept the lot.
2. Shows probability of lot acceptance P_a as function of lot quality level (p).
3. The curve starts high (close to 1.0) when the lot has very few defectives meaning the probability of acceptance is very high for good-quality lots.
4. As the defect level increases, the curve slopes downward, the chance of accepting decreases.
5. At very high defect levels, the probability of acceptance approaches zero.

This slope shows how effectively the sampling plan discriminates between good and bad lots.

If $n \gg$

1. Larger n =inspect more units, the plan becomes stricter and more discriminating (better discrimination).
2. The OC curve gets steeper means it drops sharply from high acceptance to low acceptance over a small range of defectives.

If $n \ll$

1. Smaller n =inspect fewer units, the plan is less strict, more variation.
2. The OC curve becomes flatter (poorer discrimination).
3. Good lots might get rejected more often, and bad lots may slip through more often.

Average Outgoing Quality (AOQ)

Average Outgoing Quality (AOQ) is the expected proportion of defective items in the outgoing (accepted) lots after inspection, under a specific sampling plan.

- If a lot is rejected, usually 100% inspection is carried out, and defective items are replaced with good ones before shipping.
- If a lot is accepted, it goes as it is (defective items remain in it) and any defective items in the sample will be replaced.
- This approach is called rectified inspection. So, AOQ shows the quality level of lots actually reaching the customer, considering the effect of rejection and rectification.

- *The equation for AOQ is*

$$AOQ = \frac{P_d P_a (N - n)}{N}$$

Where,

P_d = True proportion defective of the lot

P_a = Probability of accepting the lot

N = Lot size

n = Sample size

$$AOQ = \frac{P_d P_a (N - n)}{N}$$

$$= P_a \times \frac{d}{N}$$

$$= P_a \times P$$

Where,

d = Actual number of defectives in the lot

$\frac{d}{N}$ = p , actual lot defective fraction

If given, Lot size (N)=1000

Actual defectives in the lot (d)=40

Sample size (n)=50

Acceptance number (c) ≤ 2

Probability of accepting the lot, $P_a = \frac{13}{20}$

So, defect fraction (p)=40/1000

$$= \mathbf{0.04}$$

$$= 4\%$$

So, AOQ= $P_a \times P$

$$= 0.65 \times 0.04$$

AOQ = 2.6% defective items in outgoing lots